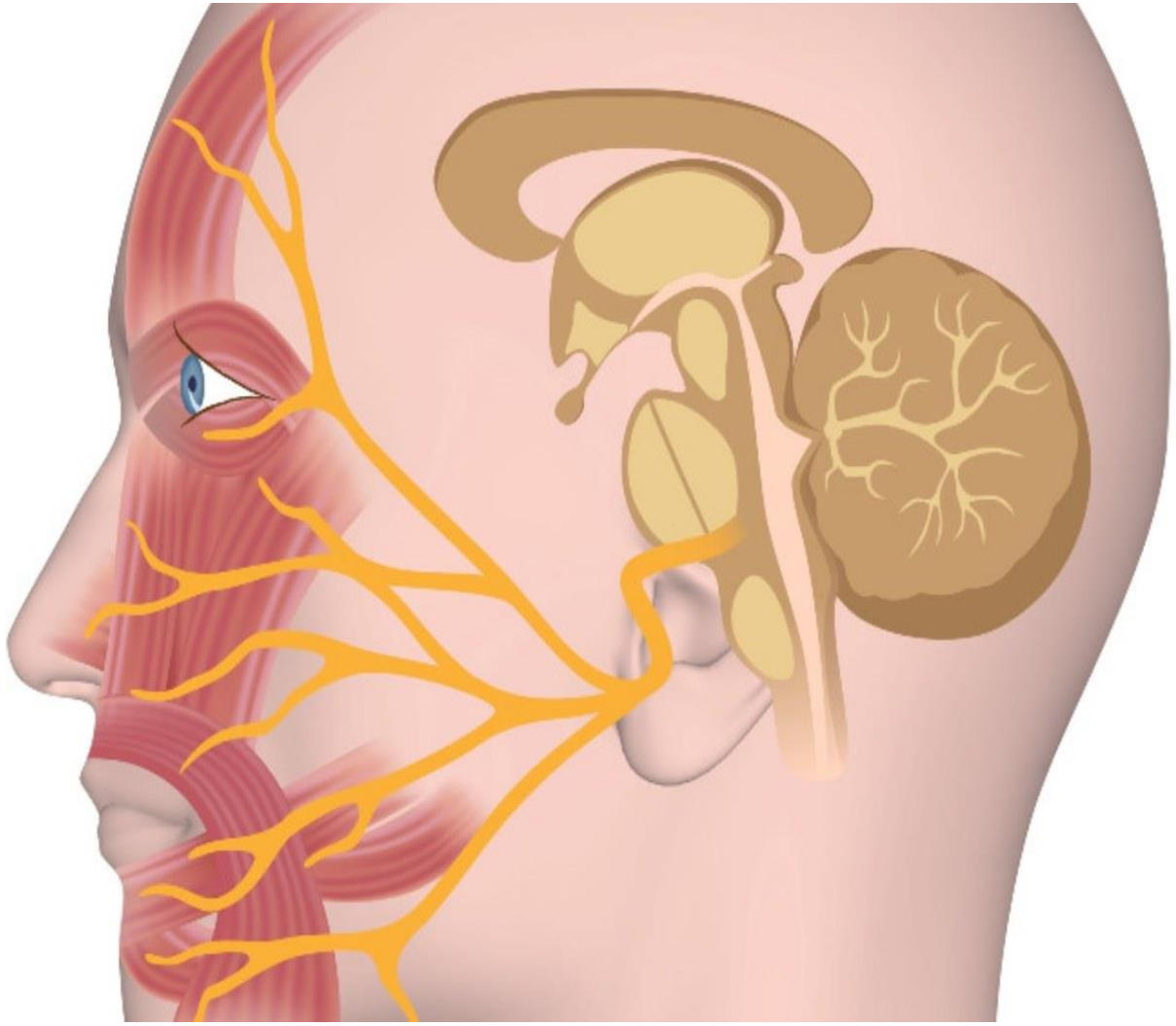
# Physiology of Mouth and Esophagus

MEDICAL PHYSIOLOGY

# ILOs

1. Understand chewing reflex.
2. Name salivary glands, mechanism and control of salivary secretion
3. Describe  functions of saliva
4. Explore digestive functions of the saliva and phases of salivary secretion.
5. Describe different phases of swallowing
6. Explore protective reflexes during swallowing.

# Mastication (chewing)

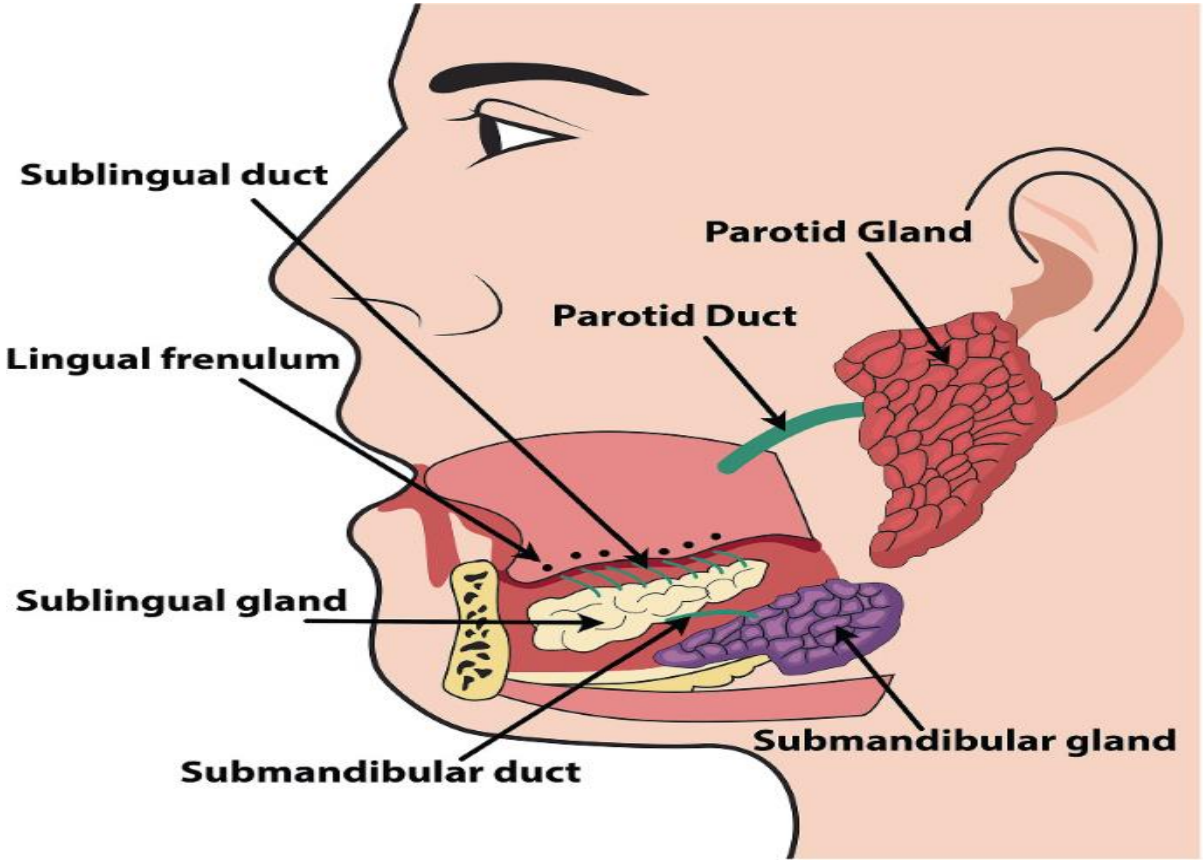


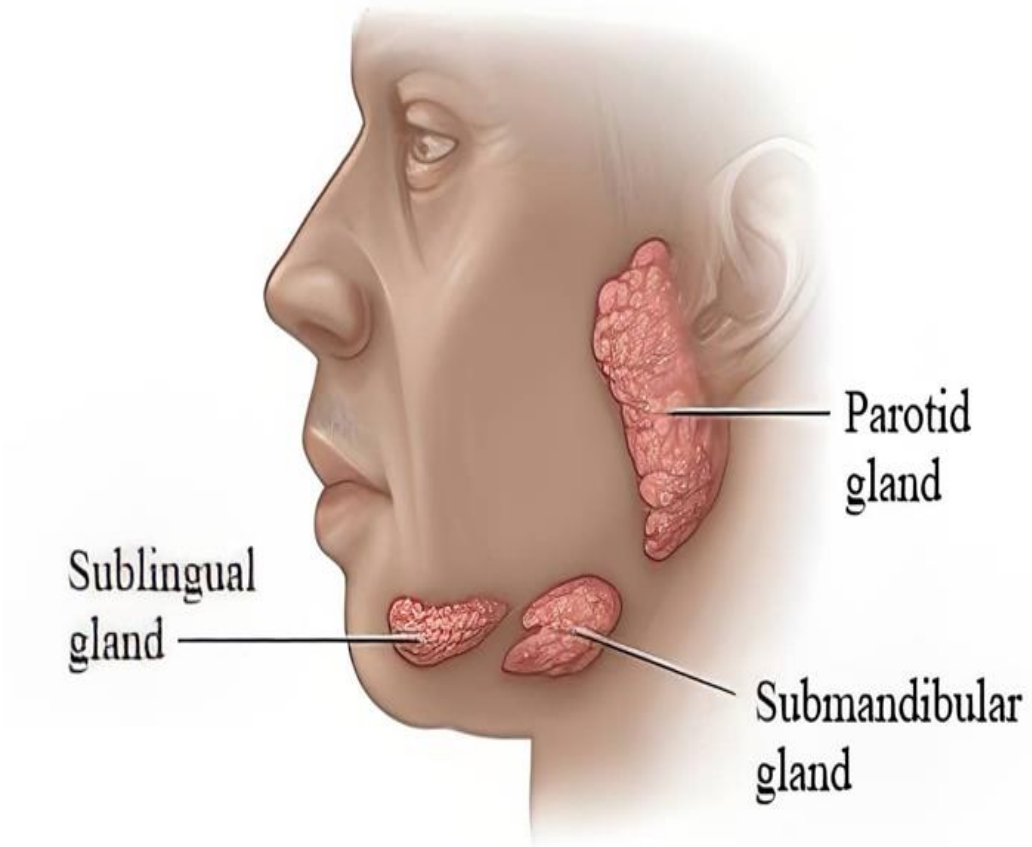
# Mastication (chewing)

## It is rhythmic opening & closure of mouth

- **Muscles** of chewing are innervated by the fifth cranial nerve ($5^{th}$ = trigeminal nerve)
  They are controlled by nuclei in the brain stem
- **Chewing reflex**:
  - The presence of a bolus of food in the mouth causes reflex inhibition of the muscles of mastication & dropping of the lower jaw
  - The drop, in turn, initiates a **stretch reflex** of the jaw muscles that leads to contraction. This is repeated rhythmically
- **Importance**:
  - ❶ It breaks large food particles into smaller ones
  - ❷ Stimulates: 1-**Taste** buds
    2-**Smell** receptors leading to sense of satiety
  - ❸ Stimulates salivary secretion which helps swallowing process

# Salivary glands



# Salivary glands & Saliva

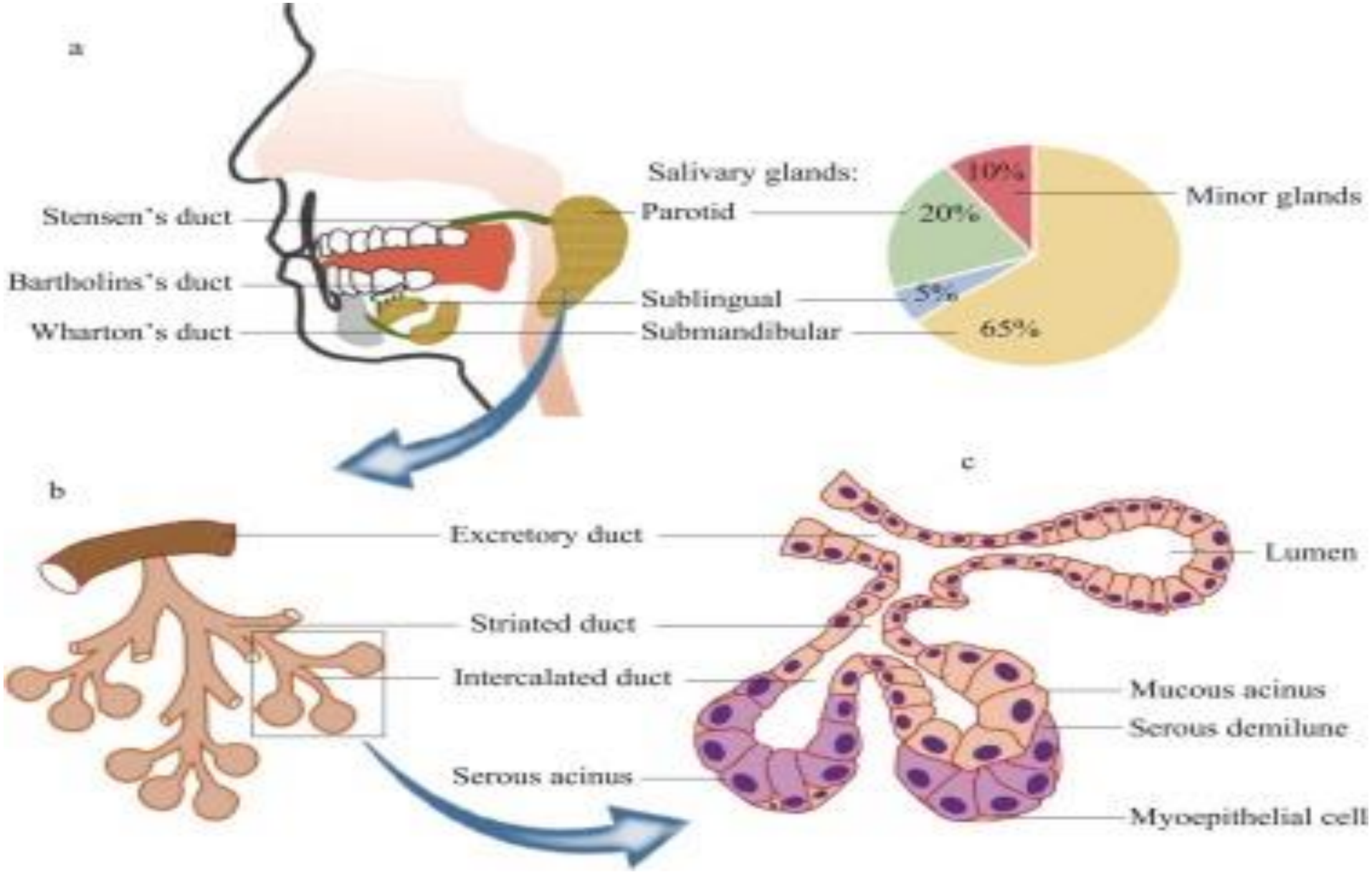
## ⋆ Types of salivary Secretory cells:

- There are two types of salivary Secretory cells:

❶ Serous cells: Secretes serous (watery) secretion containing ptyalin (α amylase enzyme)

❷ Mucous cells: Secretes mucous (viscous) secretion containing mucin

1. Parotid glands acini are **serous** & form 25% of saliva
2. Sublingual glands acini are **mucous** & form 5 % of saliva
3. Submandibular glands acini are **mixed** & form 70% of saliva
4. Buccal glands secretes only **mucous**


a
Stensen's duct
Bartholins's duct
Wharton's duct
Salivary glands:
Parotid
Sublingual
Submandibular
10%
20%
5%
65%
Minor glands
b
Excretory duct
Striated duct
Intercalated duct
Serous acinus
c
Lumen
Mucous acinus
Serous demilune
Myoepithelial cell

★ **<u>Composition of saliva</u>:**

1-Volume: 1500 ml/day

2-pH: 7

3-Contents: ❶Digestive enzymes: Ptyalin, lingual lipase & mucin

❷Electrolytes: Potassium & Sodium, chloride, bicarbonate

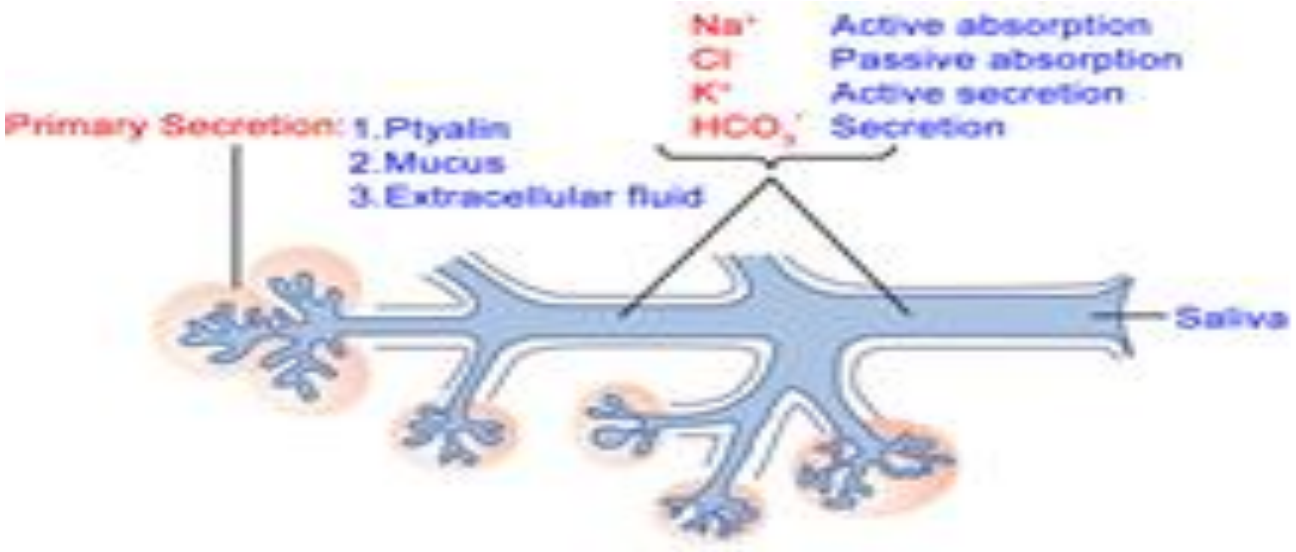
❸Saliva is hypotonic to plasma:

Contains:

- Less $Na^+$
- Less $Cl^-$
- More $K^+$
- More $HCO_3$

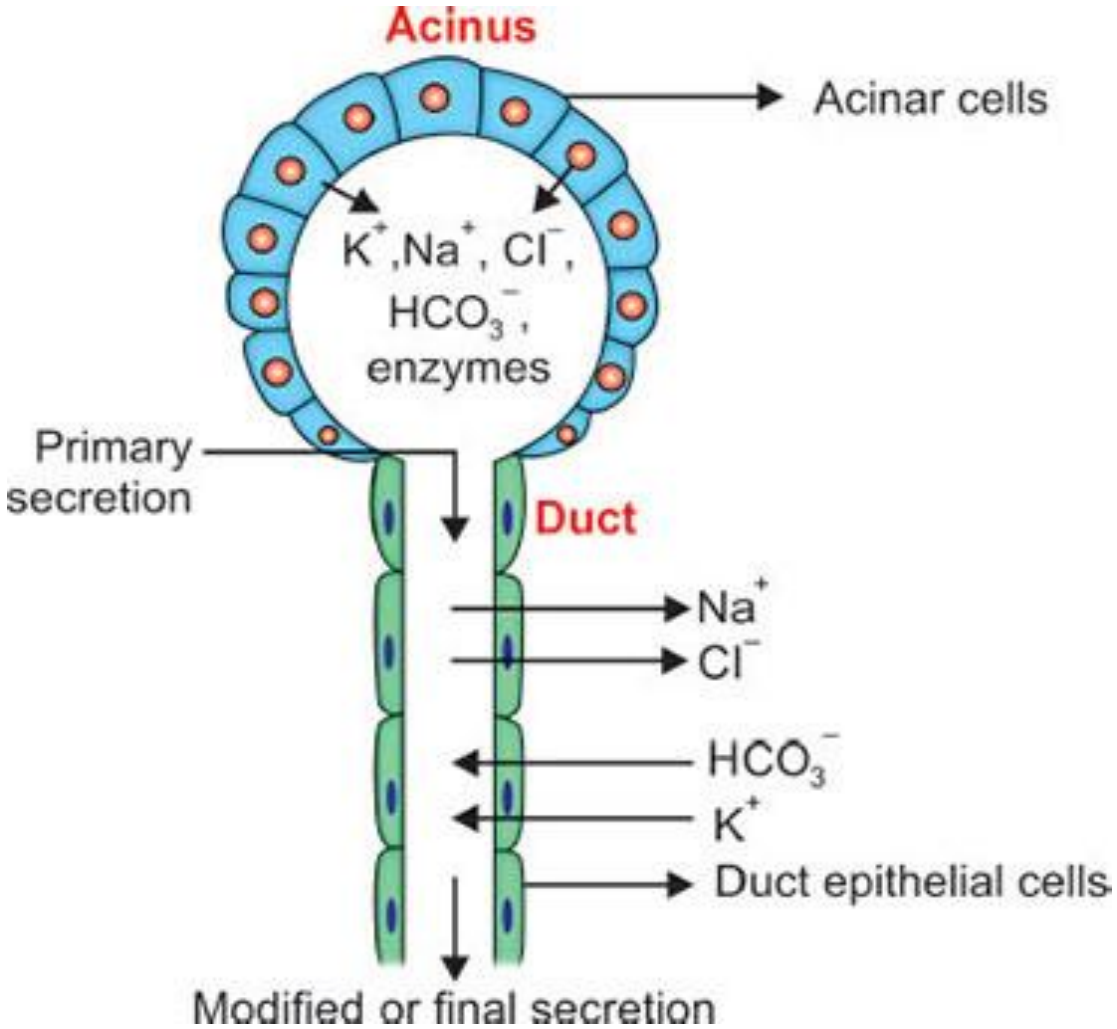
❹Immunoglobulin A (IgA): To defense against bacteria & virus

❺Lysozyme: Which attack bacterial wall




**Formation and secretion of saliva by a submandibular salivary gland**

# Composition of saliva:



★ **Functions of saliva:**

1-Protection of oral mucosa:

1. Cooling hot foods
2. Cleaning the mouth: 1]Through washing pathogenic bacteria &
   2]Their destruction by lysozyme and antibodies
3. Acid neutralization: This relieves heart burn if gastric juice regurgitated to esophagus

Saliva contains bicarbonate & mucin, which maintain oral pH at 7.

- At this pH, the saliva is saturated with calcium so teeth do not lose calcium to oral fluid
- Loss of calcium from teeth leads to dental caries

2-Digestion: - Saliva contains ptyalin (salivary amylase) which digests starches into disaccharides.
- It is inactivated in stomach & reactivated in small intestine

3-Deglutition: Saliva lubricates food because of its mucin

4-Speech: Keeps mouth moist facilitating movements of lips & tongue

5-Solvent: For food molecules to stimulate taste buds

★ **Mechanism of salivary secretion:**

It is two stages: 1-First stage:

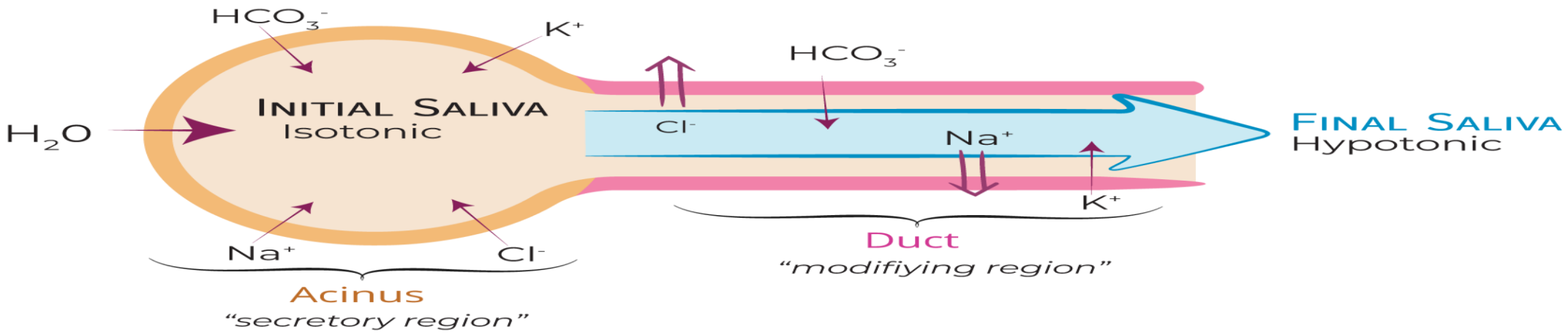
The **acini** secrete a primary secretion which contains ptyalin & or Mucin in a solution which has the same ionic composition as plasma ($Na^+$, $K^+$, $Cl^-$ & $HCO_3$) ie primary secretion is **isotonic**.

2-Second stage:

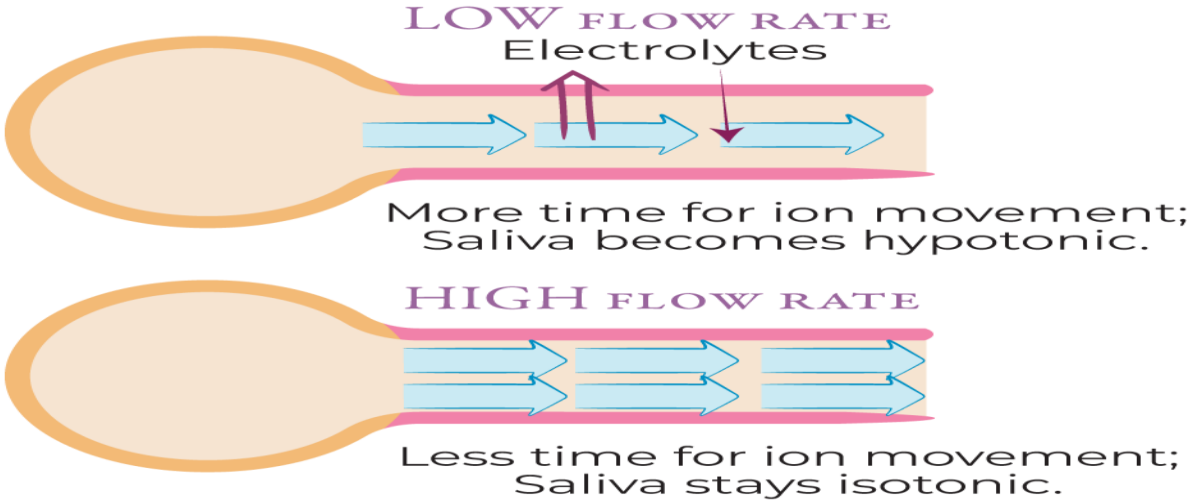
- The ducts modify the primary salivary secretion.
- $Na^+$ is actively reabsorbed in exchange with $K^+$ which is actively secreted.
- This exchange is stimulated by **aldosterone**
- $Na^+$ reabsorption exceeds $K^+$ secretion & this causes $Cl^-$ to be reabsorbed passively ie $Cl^-$ ions follow $Na^+$ ions. ie secondary secretion is **hypotonic**
- Bicarbonate ions are actively secreted into the ducts
- The net result under resting condition:
  - Na, Cl is **less** in saliva while
  - K, $HC_3$ are much **more** than the plasma

# Saliva

## Saliva Formation




## Flow Rate Effect on Composition



★ **<u>Control of salivary Secretion</u>:**

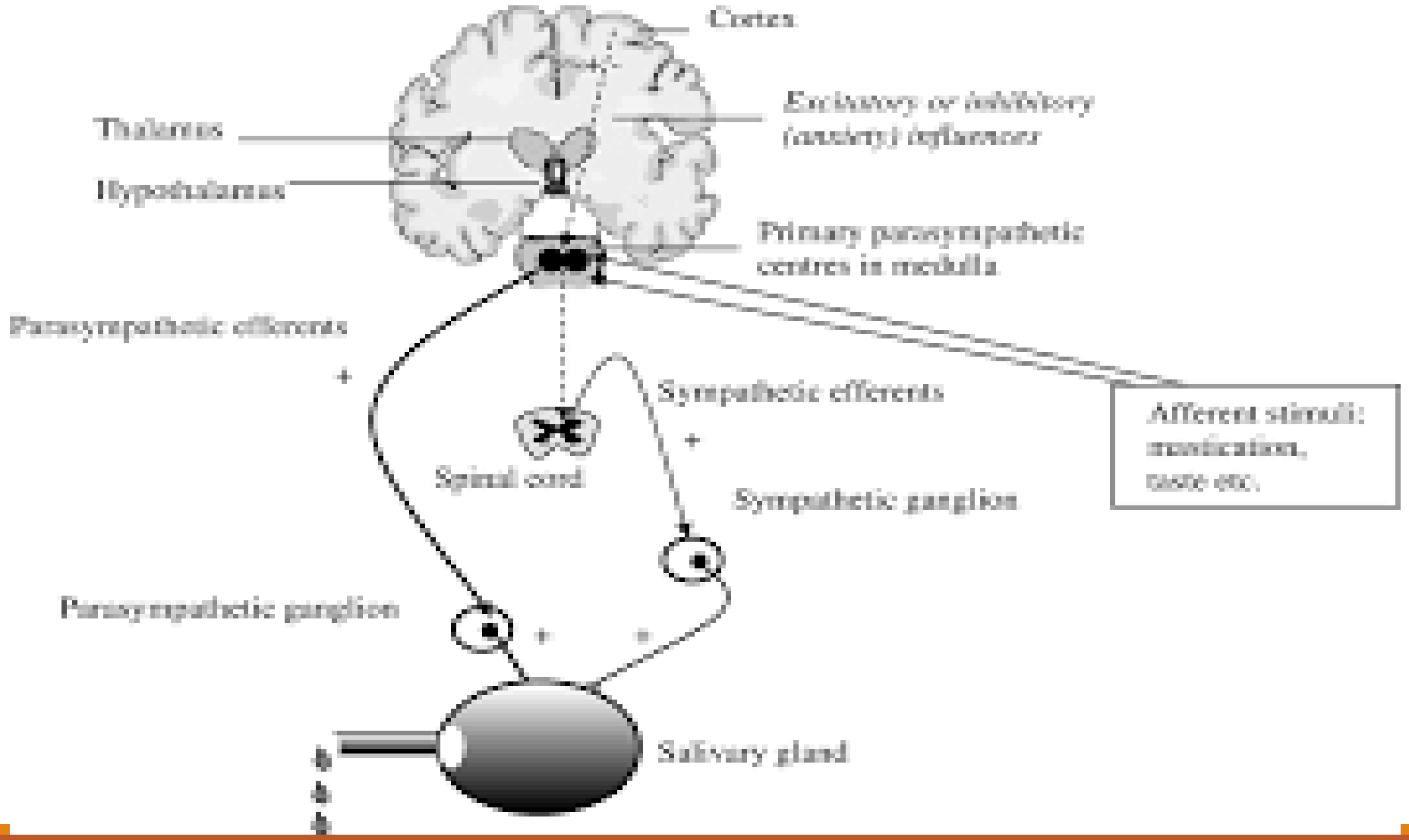
- It is under nervous control only (through autonomic nervous system)
- **Secretion of saliva can occur in response to**:

1- **Conditioned reflexes**:
- They are acquired reflexes
- They need:
  - Training (past- experience)
  - Intact cerebral cortex

a) **Cerebral cortex**: Sight, smell, hearing the preparation of food or thinking of food stimulate the cerebral cortex (because all sensations must reach the cerebral cortex) which in turn stimulates the salivary nuclei leading to salivation

b) **Hypothalamus**: There is:
- Appetite center in the hypothalamus which receives signals from the taste and Smell area in the cerebral cortex which in turn stimulates the salivary nuclei.

# Control of Salivary Secretion:



1-**Unconditioned reflexes**: They are inborn reflexes

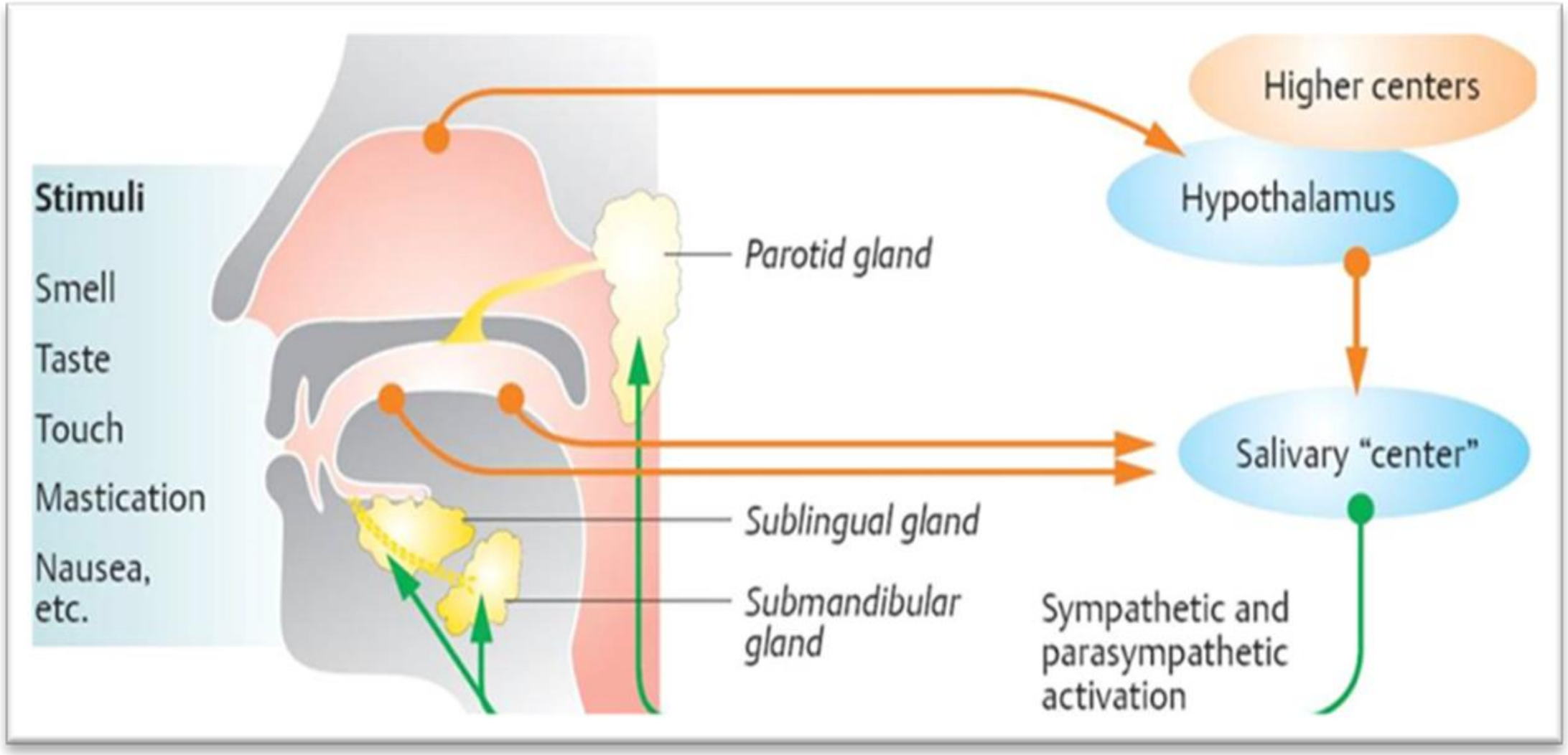
2-**The receptors are**:-Taste-thermo receptors
-Touch receptors in mouth
-Lower esophagus
-Stomach and upper small intestine

NB: Acidic lemon is the most powerful stimulus to taste

-**The afferents are**: Facial, gloss pharyngeal, trigeminal and vagus nerves.

-**The center**: Salivary nuclei

-**The efferent**:Facial & gloss pharyngeal nerves resulting in reflex increase in salivary secretion


Higher centers
Hypothalamus
Stimuli
Smell
Taste
Touch
Mastication
Nausea, etc.
Parotid gland
Salivary "center"
Sublingual gland
Submandibular gland
Sympathetic and parasympathetic activation

# Esophagus

1-The esophagus is: -Muscular tube 20 cm long

-The upper ⅓ is **striated** muscle

-The lower ⅓ is smooth muscle

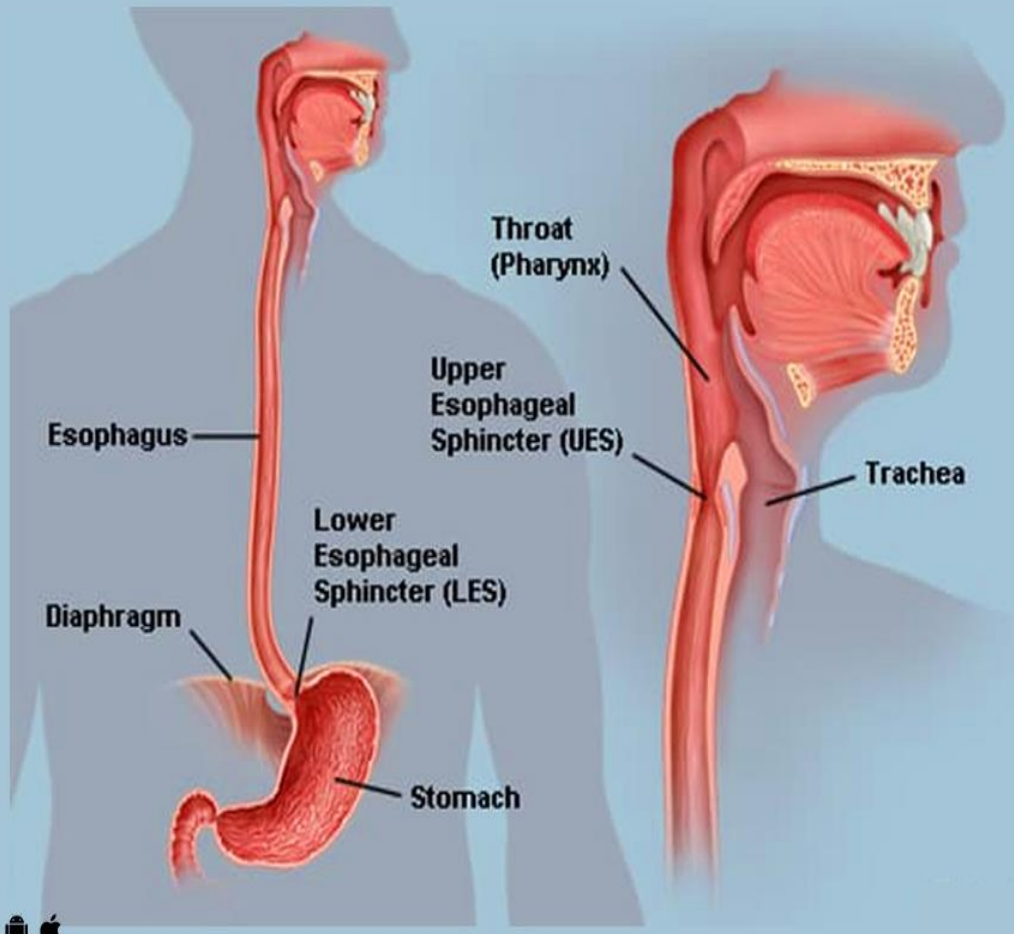
-The middle ⅓ is mixed.

2-The ends of the esophagus are closed by:

-The upper end is **closed by** the Upper Esophageal Sphincter (UES) (pharyngeoesophageal sphincter)

-The lower end is **closed by** the Lower Esophageal Sphincter (LES)

# Esophagus

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# Swallowing (Deglutition)

★ **It is initiated** voluntary & **completed** involuntary.

★ **Center**: Medulla

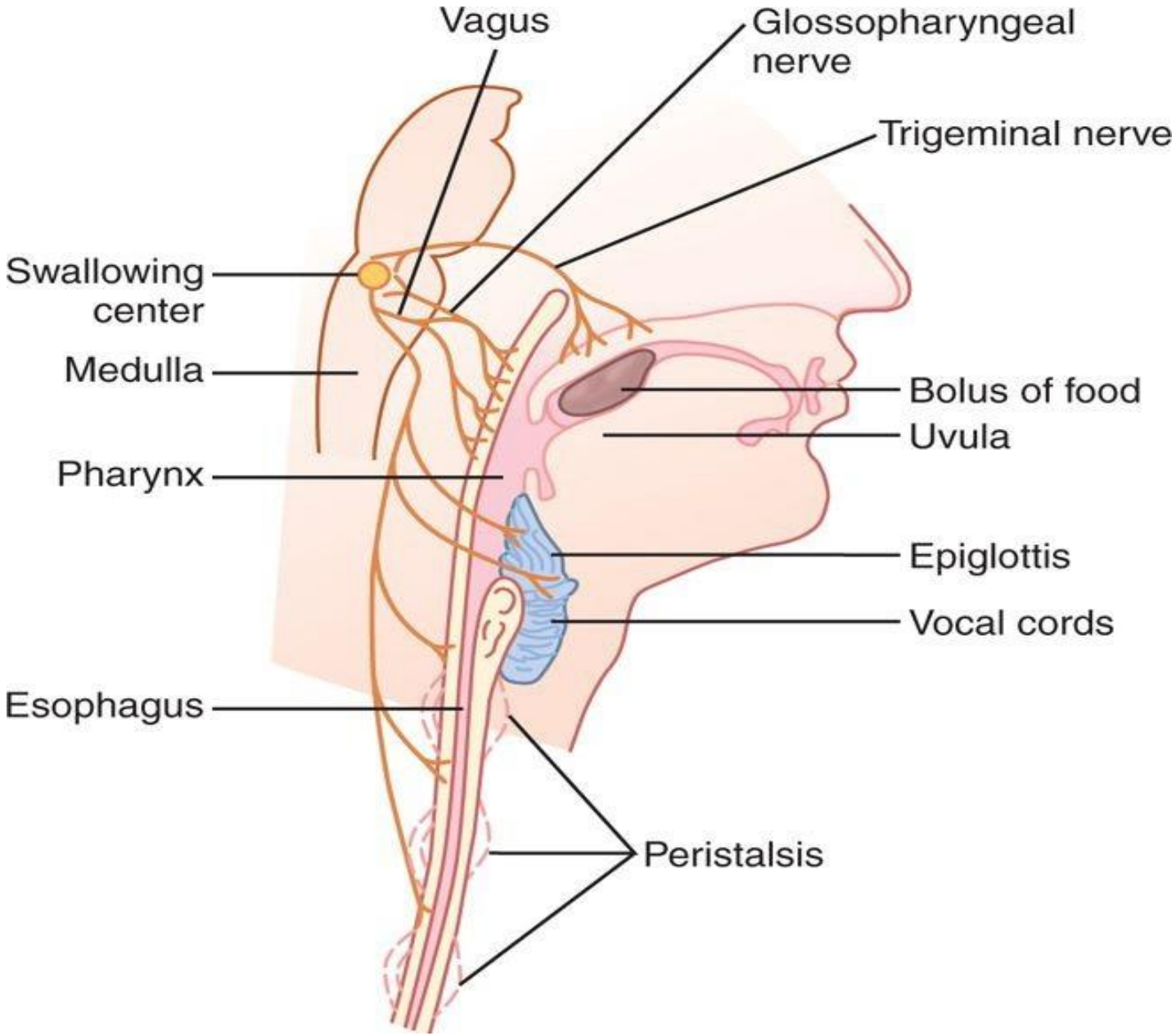
★ **Stages (Phases):**

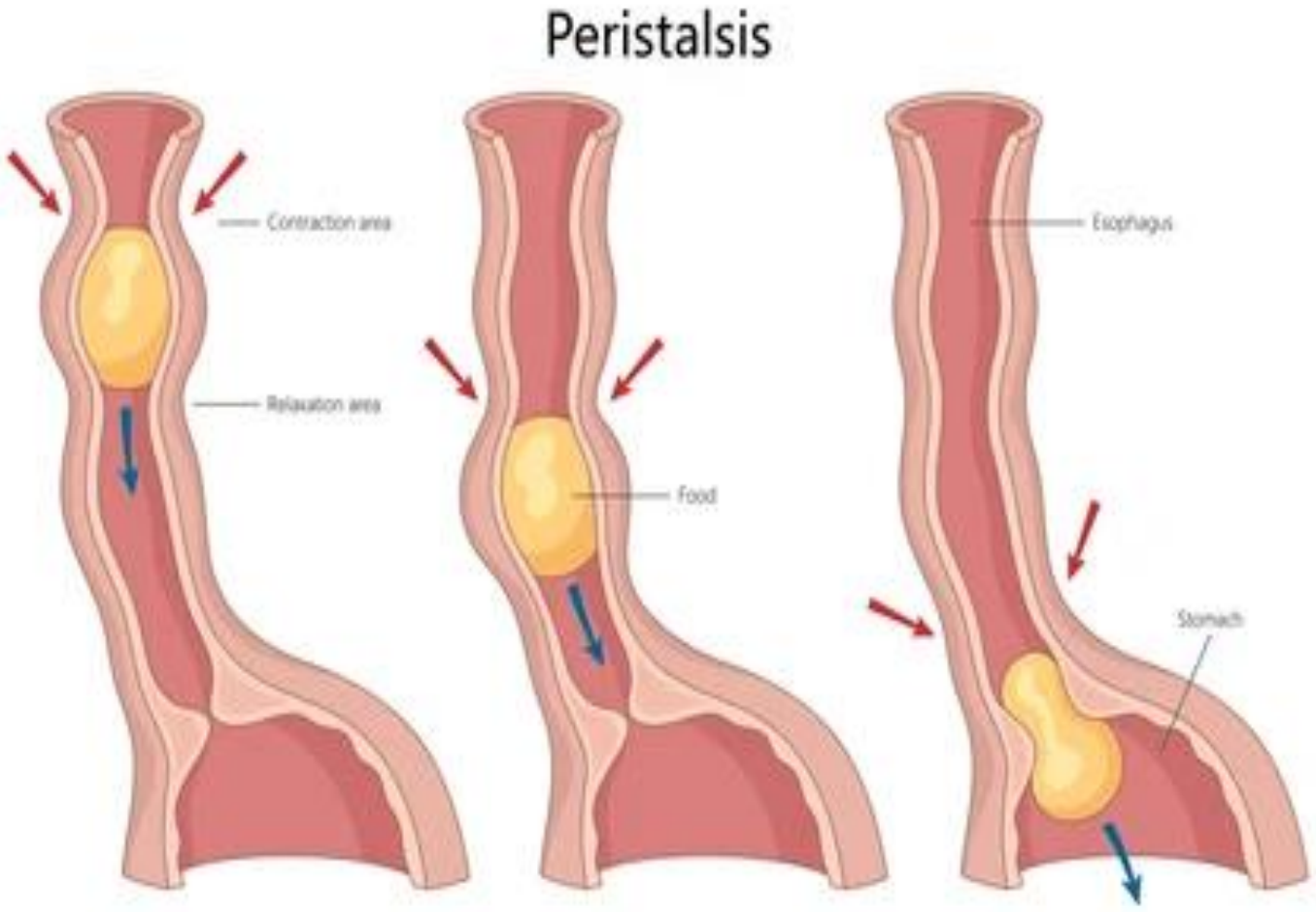
❶ Buccal phase: *(It is Voluntary)*

*(It is passage of food from mouth to pharynx)*

1-The front of the tongue is:
- Elevated & pressed against hard palate
- So bolus of food is rolled backwards towards the back of the tongue

2-Bolus is forced into pharynx by: Contraction of **mylohyoid muscle** (skeletal muscle in back of throat)


Vagus
Glossopharyngeal nerve
Trigeminal nerve
Swallowing center
Medulla
Bolus of food
Uvula
Pharynx
Epiglottis
Vocal cords
Esophagus
Peristalsis


Peristalsis
Contraction area
Relaxation area
Food
Esophagus
Stomach
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❷ Pharyngeal Phase: *(It is Involuntary)*

*(It is passage of food from mouth into esophagus through pharynx)*

In 1 to 2 seconds:

1-Presence of bolus of food in pharynx **stimulates receptors** around its opening

2-**Afferent** impulses: - Are carried along **trigeminal** & **gloss pharyngeal nerves**

- These impulses are integrated in medulla (the **center**)

3-**Efferent** impulses:

- Are carried along: 1-Trigeminal ($5^{th}$ CN)

2-Facial ($7^{th}$ CN)

3-Glossopharyngea1($9^{th}$ CN)

4-Hypoglossal ($12^{th}$ CN)

- These impulses **result in**:

  Successive contractions of **superior**, **middle** & **inferior** pharyngeal constrictors pushing the bolus of food to esophagus [as the upper esophageal sphincter (UES) relaxes]

- The bolus is **prevented from passing to: (protective mechanisms)**
  1. Nose which is **closed By** elevation of soft palate
  2. Mouth which is **closed by** elevation of:
     1] Tongue
     2] Contraction of mylohyoid muscle
  3. Larynx which is **closed by**:
     1] Elevation of larynx to **be covered by epiglottis**
     2] Closure of **glottis** (approximation of vocal cords)
     3] Inhibition of respiration (swallowing apnea)

❸ <u>Esophageal Phase</u>: *(It is Involuntary)*

*(It is passage of food from pharynx to stomach through esophagus)*

1-The Upper Esophageal Sphincter (UES):

- **Prevents air** from entering the stomach during inspiration.
- It relaxes during the **passage of food**, then contracts

2-Travelling along the esophagus:

1. Fluid: Travels by **gravity**
2. Bolus of food:

❶ Primary peristalsis:

It is: - Continuation of **peristalsis in pharynx**

- Duration: Is 8 to 10 sec (However; food takes only **5- 8 sec** to reach the stomach due to effect of gravity)

❷ Secondary peristalsis:

It: - Occurs if primary one fails

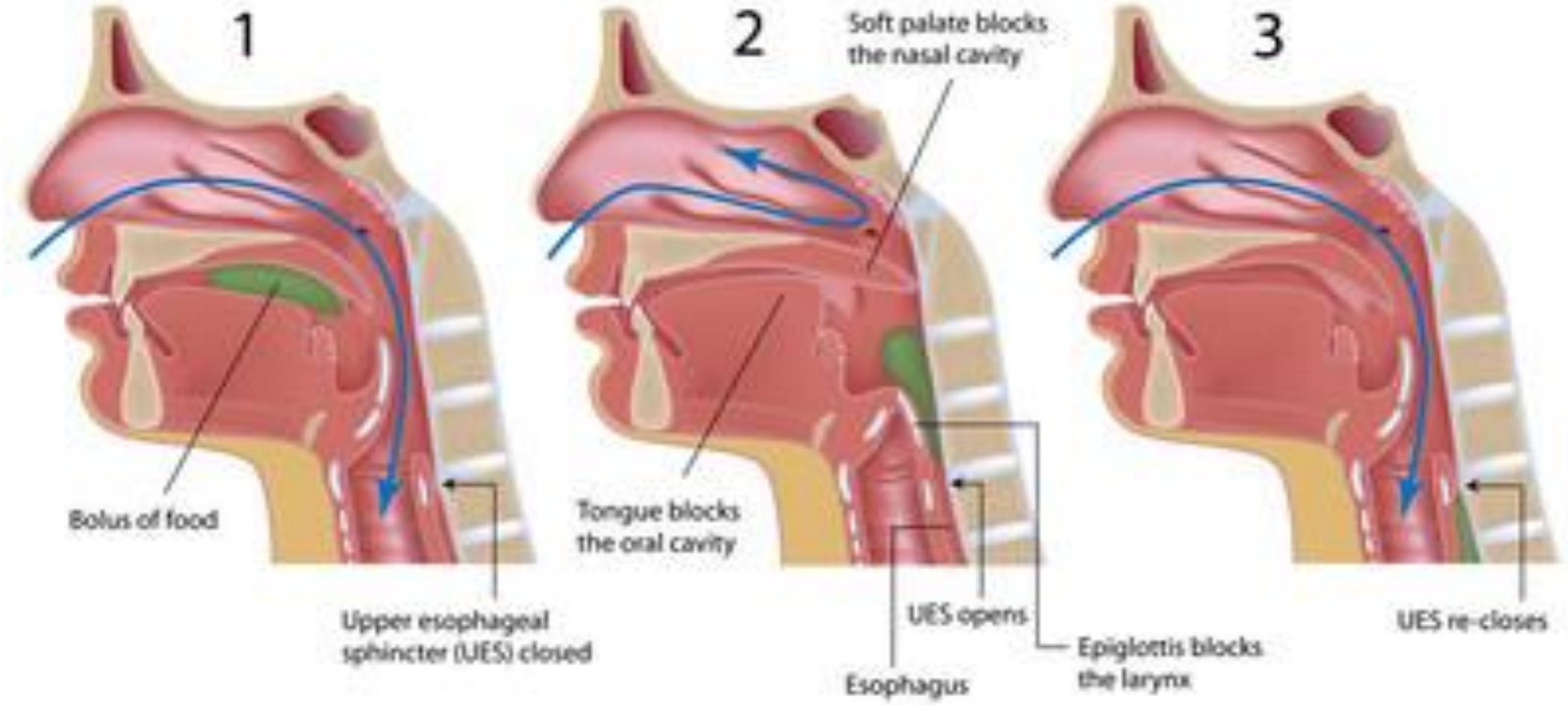
- Initiated by: - Enteric nervous system in esophagus
  - Reflexly through **vagovagal** ie:
    - Afferent is: Vagus
    - Efferent is: Vagus

1. Lower esophageal sphincter (LES) (gastro esophageal sphincter)

   During swallowing, the LES relaxes to allow food passage

NB: LES is:
- A physiologic sphincter
- Tonically constricted
- Prevents the regurgitation of gastric HCl to esophagus to prevent heart burn

# Swallowing


1
2
3
Soft palate blocks the nasal cavity
Bolus of food
Tongue blocks the oral cavity
Upper esophageal sphincter (UES) closed
UES opens
Epiglottis blocks the larynx
Esophagus
UES re-closes

Thank You